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MILITARY SPECIFICATION

DISTRIBUTORS, LIQUID BITUMINOUS MATERIAL, TANKLESS-TYPE, TRAILER-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of trailer-mounted, bituminous-material distributors with auxiliary burner, pump, and attachments.

1.2 Classification. The bituminous distributors shall be of the following types, as specified:

Type I - With spraybar.
Type II - Without spraybar.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-771	- Distributors, Bitumen and Water, Packaging of.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-S-8660	- Silicone Compound.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy-Duty, Single-Conductor and Multiconductor.
MIL-C-13486/12	- Cable, Special Purpose, Electrical, Low Tension, Heavy Duty Multiple Conductors Shielded.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-R-46164	- Rustproofing for Military Vehicles and Trailers.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-209	- Slinging and Tiedown Provision for Lifting and Tying Down Military Equipment.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Material.
MS27142	- Connector, Plug, Electrical - Pin Contact No. 12, 14, and 16 A.W.G., Waterproof.

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MS27143	- Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical: Socket Contact, No. 14 and 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way, No. 12, 14, and 16 A.W.G., Waterproof.
MS39153	- Coupling, Quick Disconnect, 3- and 4-Inch, for Construction Equipment.
MS53004	- Valve, Relay, Emergency-Breakaway, Air Brake.
MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

2.1.2 Other Government documents, and publications. The following other Government documents, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the U.S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications and standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

ANSI/AWS D1.1 - Structural Welding Code - Steel.
ANSI/AWS D1.2 - Structural Welding Code - Aluminum.
AWS B2.1 - Standard for Welding Procedure and Performance Qualifications.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95804.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

TTMA Recommended Practices.
TTMA Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 1616 P Street NW, Washington, DC 20036.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

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(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC.

Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, NY 10017.)

(Nongovenment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The distributors shall consist essentially of a gasoline or diesel engine, bitumen transfer pump, and metering device mounted on a two-wheel pneumatic-tired trailer. All controls, accessories, and other components necessary for all operations specified herein shall be provided. The type I distributor shall have a full circulating spraybar, and the type II distributor shall be furnished without spraybar. The trailer chassis shall provide mobility for towing the distributor by truck on graveled roads and paved highways at speeds specified herein, without any damage or permanent deformation to the distributor, bottoming of the suspension, or loosening of any components. Except as specified herein, the distributor automotive type components, assemblies, and accessories shall be construed as “commercial automotive products”, and they shall comply with all Federal Motor Vehicle Safety Standards applicable on the date of manufacture to nonmilitary, nonexport motor vehicle product.

3.1.1 Weights and dimensions. The gross vehicle weight (GVW) of each distributor shall include the weight of the distributor in operating condition with all attachments, equipment, and accessories. The physical characteristics of each distributor shall be as specified in table I, when the distributor is in travel position on level ground, and the inflation pressure of the tires is as recommended in the T&RA Yearbook and Supplementary Service Data Book for GVW operation at highway speeds.

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TABLE I. Weights and dimensions.

Characteristics	Requirements
GVW rating, maximum.	5,000 pounds
GVW distribution on towbar eyes.	5 to 10 percent of GVW
Overall width, maximum.	96 inches
Overall height, maximum.	96 inches
Ground clearance, excluding running gear, minimum	12 inches
Towbar height range (SAE J849).	31 through 37 inches
Angle of departure, minimum.	25 degrees

3.1.2 Compatibility. The distributor shall be compatible for towing behind a 5 ton, 6 x 6, military design dump truck.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of distributors from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish distributors conforming to this specification.

3.3 Materials. Materials shall be as specified herein. Materials not as specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The distributor shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant to the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components or sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the distributor may be newly fabricated from recovered materials to the

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maximum extent practicable, provided the distributor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the distributor.

3.4 Trailer. The distributor shall be mounted on a cart-type trailer as specified herein, and shall provide the on-highway and secondary road mobility as specified.

3.4.1 Mobility. The distributor shall perform as specified herein when: (1) equipped with all attachments, equipment and accessories stowed for travel; (2) towed over highways, secondary roads, and rough unimproved roads by the truck specified in 3.1.2.

3.4.1.1 Highway travel. The distributor shall travel over smooth, hard surfaced, primary highways at sustained speeds up to 55 miles per hour (mph).

3.4.1.2 Secondary road travel. The distributor shall travel over gravel and improved roads at speeds up to 30 mph.

3.4.2 Trailing and turning ability. The distributor shall trail the towing vehicle without bottoming, cramping, weaving, or sidesway. The distributor shall follow the truck without exceeding the tracking deviation, as specified in 393.70 of the DOT Motor Carrier Safety Regulations, when towed over a paved surface. Jackknife turns of at least 60 degrees, as measured from straight ahead to the right and to the left, shall not cause damage to or interference between the truck and distributor.

3.4.3 Frame. The frame shall be of the ladder type with longitudinal frame side rails reinforced with transverse crossmembers and gussets to meet the transport and the equipment operating conditions. The web of the side rails shall be continuous from the towbar to the last crossmember. Gusseted crossmembers shall tie together, maintain alinement, and distribute the stress of the suspension mounts.

3.4.4 Suspension. The distributor shall be equipped with a sprung, single axle suspension system, tailored to the distributor. The suspension manufacturer's rated capacity shall be not less than the loading imposed on each member, measured at the ground, with the distributor loaded to the maximum GVW. The suspension shall provide a nondamaging ride on graveled surface secondary roads and a safe, stable ride on paved highways. The suspension shall have elastomer bushing or blocks that eliminate the need for lubrication. The installation of the suspension shall be in correct alinement with the frame and with the axle so that the two diagonal measurements from the center of the lunette to the outer center of each spindle on the axle do not differ in excess of 1/16-inch tolerance for every 20 feet or fraction thereof in diagonal distance. Elastomer cushioned bumper stops or blocks shall be furnished to prevent axle-to-frame contact and limit suspension articulation at full compression (jounce) and rebound. Shock absorbers shall be furnished to control jounce and rebound.

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3.4.5 Axle assembly. The axle shall be rated by the axle manufacturer at a continuous load capacity rating of not less than 10,000 pounds for commercial highway trailer service. The spindles shall be alloy steel forgings, correctly hardened and ground as recommended by the spindle manufacturer, and shall be located in the correct alinement position with the axle tube. The spindles shall be equipped with assemblies of hubs, bearings, seals, and foundation brakes of the type specified herein. Hub assemblies shall be equipped with semitrailer industry standard tapered roller bearings having a standard inside-to-inside bearing spacing. Cups, bearings, and cones of all bearings shall be in matched sets from the same bearing manufacturer. The axle hubs shall conform to SAE J694.

3.4.6 Wheels, rims, and tires. The distributor shall be equipped with single demountable disc wheel and tire assemblies. The disc wheels shall be of the heavy duty type with T&RA rim. The rim and wheel disc shall be constructed by the same wheel manufacturer and shall be in accordance with the wheel manufacturer's recommendations for the loads and service specified herein. Rims, tires, inner tubes and flaps (if furnished), valve cores, and caps shall be in accordance with the T&RA Yearbook and Supplementary Service Data Book. The tires shall be of not less than 100 level quality and shall be of the highway or traction lug type. All tires shall be of the same size and load range and shall not be more than 12 months old on the date of shipment to the Government. The tires shall be the tube or tubeless type, having a minimum size of 9.00 – 20, load range (10 ply rating). The load imposed on each tire, measured on level ground under loaded travel conditions, shall not exceed the recommended rated loads and inflation pressures of the tires for 50 mph highway speed for continuous highway operation. Valve caps shall be of metal and shall be equipped with elastomer inner seals. Valves shall be of the heat resistant type. Tire inflation pressures recommended by the T&RA shall be stencilled in l-inch high letters.

3.4.6.1 Balancing. Tires, wheels, hubs, and brake drums shall be inherently in-balance, singly and collectively after assembly, to preclude wheel tramp, drumming, or excessive tire wear, when operated up to the maximum highway speeds specified herein.

3.4.6.2 Tire carrier. When specified (see 6.2), one tire carrier shall be provided and installed on the curb side on the front of the distributor. Means shall be provided for securing and locking the tire assembly in the carrier preventing accidental loss or theft.

3.4.6.3 Spare tire assembly. When specified (see 6.2), one inflated spare tire and wheel identical to those furnished on the distributor shall be installed in or mounted on the tire carrier.

3.4.7 Fenders and mudguards. The distributor shall be equipped with two fenders and two rear splash guards. The fenders and splash guards shall be in accordance with SAE J321, SAE J682, and SAE J683. The fenders and splash guards shall restrict water and other road surface substances thrown from the distributor wheels. The fenders either shall be contoured to the normal axle centers or shall be octagonal shape about the wheels and shall include inner splash shields. The fenders shall support a weight of not less than 200 pounds imposed on any

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l-square-foot area at any one time without damage or permanent deformation. The fenders shall be constructed of lightweight fiber glass reinforced plastic or aluminum material. The top surface of the fenders shall be treated with an antiskid surface. The outer edges of the fenders shall be rolled to reduce side road splash. The splash guards shall be of a size not less than 12 by 18 inches, and shall include antisail characteristics. The guards shall be of fiber glass reinforced polyester or of solid polyvinyl chloride homopolymers and monomeric low temperature plasticisers without reinforcement.

3.4.8 Rear bumper. The distributor shall be equipped with a horizontal rear bumper, extending the full width of the distributor. The bumper shall be not less than 8 inches vertical height and shall extend rearward not less than 2 inches from any part of the distributor. The left and right ends of the bumper shall be rounded and closed to prevent hooking when sideswiped by an obstruction.

3.4.9 Intervehicular connections. The distributor shall be furnished with an intervehicular electrical harness and two airbrake lines for connecting the truck's electrical and airbrake systems to the distributor systems. All intervehicular connections shall be installed on the distributor so that they extend forward between 36 and 42 inches toward the towing vehicle, and rearward at least 20 inches on the towbar, measured from the center of the towbar eye. A spring clip shall be installed to hold the electrical harness plug to the towbar with the harness doubled back on the distributor. Two dummy couplings shall be installed to hold the airbrake line ends with the lines doubled back. Each dummy coupling shall be mounted perpendicular to the mounting surface and clearance shall be provided for the spring dust cover on the coupling. Intervehicular connections, when connected and when stowed, shall be routed or protected so that dropping of the tailgate of the towing vehicle shall not cause physical damage.

3.4.10 Towbar components. A towbar eye conforming to SAE J847 shall be furnished and installed on the towbar within 1/8-inch tolerance of the longitudinal centerline of the distributor. The towbar shall be within the height range specified in table I. The lunette (towbar eye) shall be inserted in a support that is welded into the towbar. The lunette support shall prevent lunette rotation. Two safety chains, equipped with safety type grabhooks, shall be furnished and shall conform to SAE J697 and to the DOT Motor Carrier Safety Regulations. The forward length of each chain hook shall be between 36 and 42 inches measured from the center of the towbar eye. The safety chain hook shall pass through the shackles of the towing vehicle specified in 3.1.2. The attachment end of the safety chains shall be either perimeter welded or bolted to the towbar. The towbar shall be of the A-frame type. The towbar shall be constructed to embody structural strength to meet or exceed the following requirements: support and transport the weight of the distributor at the GVW specified in table I; transmit and distribute the imposed load to the suspension system and eye, and resist bending and twisting without permanent deformation or part failure when subjected to the performance requirements specified herein.

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3.4.11 Landing gear. A single-leg landing gear shall be installed on the towbar of the distributor. The landing gear shall be vertically adjustable for not less than 12 inches. The landing gear shall either retract or be hinged and fold to provide not less than the ground clearance of the first trailer crossmember. The landing gear shall be arranged for positive locking into park and travel positions. A folding-type crank handle for operating the vertical adjustment mechanism shall be furnished. A travel retainer for the handle shall be provided. The tangential driver's effort at the handgrip to raise the maximum GVW load specified in table I shall not exceed 30 pounds. The mechanism shall be protected from exposure to foreign matter which could impair functional efficiency. The landing gear shall incorporate a self-alining land pad which will provide a ground bearing load not exceeding 20 pounds per square inch when the maximum towbar weight in table I is imposed. The edges of the pad shall be turned up.

3.4.12 Support legs. Two adjustable support legs shall be furnished and shall be located at the left and right rear corners of the distributor. Each support leg shall be in accordance with the minimum requirements as specified for the landing gear in 3.4.11.

3.4.13 Brake system. The distributor shall be equipped with a water immersion resistant, straight airbrake system. The system shall conform to the DOT Motor Carrier Safety Regulations, the DOT Motor Vehicle Safety Standards, the California Vehicle Code, Division 12, and the SAE Handbook, except as otherwise specified herein. All brake system components and plumbing shall be located, installed, and protected to prevent damage or malfunction from ground obstruction, debris, and water and to obtain maximum safety, efficiency, and ease of maintenance. The brake system shall include service and emergency gladhands with dust covers, relay emergency valve, straight air cylinders or chambers, foundation brakes on each wheel, and all other components and plumbing required for a complete matched brake system. The gladhands shall conform to SAE J318. The glandhands shall be colored blue for service and red for emergency. A springloaded, metal, dust cover shall be installed on each gladhand to prevent contaminants from entering the opening. The dust cover shall not cause interference with another gladhand with dust cover installed. The gladhands shall be installed on the front of the trailer with the service on the curb side and emergency on the road side. In addition, brake line identification tags shall be furnished. The relay emergency valve shall conform to MS53004, except that the rapid bleed-down valve shall be replaced with a plug on the straight airbrakes. The relay valve shall conform to the DOT regulations. Each air reservoir shall conform to SAE J10. The service brake system shall have a total air storage capacity of not less than 2 times the minimum recommended air volume for truck-trailers in SAE J813. The reservoir shall be equipped with an automatic drain valve and a 90-degree twist drain cock. The drain cock shall be mounted on the low end of the reservoir and positioned on the curb side of the distributor, so as to be accessible to personnel from the kneeling position.

3.4.13.1 Brake drums. The brake drums shall be of the antibellmouth type J43l automotive gray cast iron rubbing surfaces, SAE G3500 or SAE G4000. The drums shall be of the heavy-weight type that will provide an adequate heat sink for use with metallic linings.

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3.4.13.2 Brake lining. The brake linings shall be of the nonasbestos and of the nonfading from heat and from water immersion type. The lining shall be of the standard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc.

3.4.13.3 Brake lines. The brake lines shall be routed within the protection of the framing members and shall be supported and secured by cushioned clamps at intervals not greater than 3 feet. At each point where the brake lines pass through metal members, the lines shall be protected by a rubber or nylon grommet and tightly secured, or a through-frame connector shall be used. The routing and connection locations shall facilitate replacement of the brake lines from bulk supply.

3.4.14 Chock blocks. A chock block conforming to figure 1 shall be provided for each wheel. The captive chains shall permit placing the chock block both to the front and to the rear of each wheel.

3.4.15 Vehicular electrical system. When specified (see 6.2), the distributor shall be equipped with a waterproof, 12-volt vehicular electrical system. Except as specified herein, the electrical system shall be in accordance with the following requirements:

a. DOT Federal Motor Vehicle Safety Standards (FMVSS).
b. DOT Federal Motor Carrier Safety Regulations (FMCSR).
c. SAE Handbook, Electrical Equipment and Lighting Section.
d. TTMA Recommended Practices RP No. 6, RP No. 9, RP No. 14, RP No. 40, and Technical Bulletin No. 65.

All components shall be provided with the maximum practical protection from damage, shock, and vibration through means of location and resilient mounts. The electrical components shall be in accordance with table II. Item quantities that are not specified shall be furnished as required (AR) for each distributor.

TABLE II. Vehicular electrical system.

Description	Requirements	Quantity Per heater
7 conductor plus and intervehicular harness assembly	SAE J560 and TTMA RP#40, 7-wire type with plated metal connectors, splash proof plug cover. SAE J1067 seven conductor jacketed cable fastened to connection box.	1 ea
Connection box	Splash proof type with stud and nut terminals.	1 ea min
Circuit breakers	SAE J553 protecting all live circuits.	AR

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TABLE II. Vehicular electrical system. (Continued)

Description	Requirements	Quantity Per heater
Wiring	Heavy duty type.	AR
Wire identification	SAE J560 or J821	AR
Wiring harness	Polyethylene electrical conduit, each end weatherproofed with silicone sealant or with one-half over-lapping turns of plastic electrical tape.	AR
Grommets	Provided where needed along the wire routing to prevent chafing.	AR
Supports	Elastomer cushioned support straps or elastomer ties with mounts; wiring support spacing 36 in max span, prevent whipping, chafing, and fatigue damage.	AR
Wiring splices	Mechanical splicing devices; interior waterproofed with silicone lube or compounds.	AR
Connectors and terminal	Waterproof types; blade type SAE J858, pin type SAE J928, and spade SAE J561. Blade and pin types fully enclosed and lubricated, prior to assembly with silicone lubricant; spade type equipped with heat shrinkable sleeves and coated with electrical varnish after assembly; all connectors and terminals plated or solder-coated to retard corrosion.	AR
Connector holder	Plastic or plastic-insulated metal spring clips; fastened to vehicle; furnished to anchor wire-to-wire connectors.	1 ea per connector
Running lamps	4 sidemarker, 4 clearance, 3 rear identification; per TTMA RP-108/1 or RP-108/3. Waterproof and shock resistant types.	11 ea min

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TABLE II. Vehicular electrical system. (Continued)

Description	Requirements	Quantity Per heater
Combination lamps	2 turn and tail outboard and 2 stop and tail inboard; per TTMA RP No. 14 and SAE J759; double contact; 3 wire type including ground wire; waterproof and shock resistant type.	4 ea min
Lamp lenses and bulbs	Heavy duty or fleet service type bulbs, bulbs mounted in vibration-protected sockets with insulators end stainless steel or rubber bulb tension devices.	AR
Reflectors	2 on each corner, round type, securely fastened by bolts, screws or rivets.	8 ea min
Wiring diagram	Plate furnished on trailer per TTMA Bulletin #65.	1 ea

3.4.15.1 Conversion to 24 volt electrical system. When specified (see 6.2), the distributor trailer shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact plug, the trailer shall be equipped with the following military intervehicle components:

a. Military 12-contact plug – MS75020-1.
b. Military intervehicular cable – MIL-C-13486/12 or SAE J1067 seven conductor jacketed cable.

The military plug and cable or SAE cable shall be the same length as the TTMA plug and cable and both shall be labeled with identification plates for 24-volts and 12-volts respectively, negative ground. The military 24-volt intervehicular cable shall be routed into a splashproof stud and nut type connection box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with table III.

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TABLE III. Military-commercial connections.

Function	Mil. plug terminal	Connection instructions	Comm. TTMA wiring ident.
Blackout marker.	A	No.	-
Ser. left turn & stop.	B	Connect to –	YEL (3)
Blackout marker.	C	No.	-
Ground	D	Connect to –	WHT (1)
Ser. tail & marker.	E	Connect to –	BLK (2)
		& Connect to –	BRN (6)
Blackout stop.	F	No.	-
Blackout marker.	R	No.	-
Ser. right turn & stop.	J	Connect to –	GRN (5)
Aux. power (hot).	K	No.	BLU (7)
Ground.	L	Ground to Frame	-
Electric brake.	M	No.	-
Electric brake	N	No.	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.5 Engine. The distributors shall be powered by a gasoline or diesel engine (see 6.2), having horsepower, torque and speed characteristics to satisfactorily meet all performance requirements specified herein without exceeding the engine manufacturer's intermittent rated horse power of the engine furnished. The engine's governed speed shall be set and sealed in accordance with the engine manufacturer's recommendations. The engine shall be capable of operating continuously at rated load in an ambient temperature of 120° F without exceeding the maximum allowable heat temperatures established by the engine manufacturer. The engine shall be furnished with the following accessories:

a. Fuel filter.
b. Oil filter.
c. Air cleaner, dry type with airflow restriction indicator.
d. High temperature cut off switch.

3.5.1 Engine instrument panel. The instrument panel shall be equipped with lights controlled by a weatherproof switch. The instrument panel shall be located in a position

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visible from the operator station. The instrument panel shall be equipped with the following instruments:

a. Engine lubricating oil pressure gage.
b. Engine coolant temperature indicator.
c. Engine tachometer.
d. Battery charging ammater or voltmeter.
e. Engine hourmeter.
f. Indicator, fuel quantity.

The instrument panel shall be protected from vandals by a cover plate fitted with a lock hasp that shall accommodate a padlock that has a 3/8 inch diameter shackle.

3.5.2 Engine cranking system. Unless otherwise specified (see 6.2), the engine shall be equipped with a 12 volt electrical cranking system. The electric cranking system shall include but not be limited to the following: electric starter, alternator, voltage regulator, battery and all other accessories required to complete the cranking system. The circuits shall be numbered and identified by band markers.

3.5.3 Diesel engine fuel. The diesel engine furnished shall operate on fuel oil conforming to VV-F-800, grade DF2.

3.5.4 Engine lubrication. The engine lubricating oil shall be in accordance with MIL-L-2104.

3.5.5 Engine oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. The tap shall be labeled “Engine Oil”.

3.5.6 Engine fuel tank. A fuel tank conforming to the contractor’s commercial standards, except as specified herein, shall be provided. The tank capacity shall be sufficient for 8 hours of operation at the maximum fuel rating of the engine. The tanks filler opening shall include a removable strainer and have a chain-secured filler cap. The tank shall be stenciled: “FILL WITH (specify type of fuel) FUEL ONLY”. The tank shall be provided with a fuel level indicator and with flexible fuel lines to the engine. Provisions shall be made to drain the fuel into a hand held container.

3.5.7 Batteries and battery box.

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3.5.7.1 Batteries. Unless otherwise specified (see 6.2), the distributor shall be furnished with batteries that are charged and dry.

3.5.7.2 Battery box. The batteries shall be contained in a weatherproof box fabricated of steel not less than 0.0747 inch (USS 14 gage) in thickness or in a waterproof compartment on the paver provided the compartment meets the requirements specified herein. When the batteries are installed in the box, with all electrical connections completed and with the battery-box lid in the closed position, no uninsulated electrical conductor shall be within 1 inch of any point with the battery box at top of the box, and drains shall be provided in the bottom. The box shall have built-in restraints and holddowns, constructed so that pressure will be applied to the outer battery case only. The box shall be located so that the batteries will be accessible for maintenance, servicing with a hydrometer and syringe, removal, and replacement. Grommets shall be furnished to protect the battery lead-in cables. All interior surfaces of the box, including the restraints and holddowns, shall be treated with acid-resistant paint.

3.5.8 Wiring. Unless otherwise specified (see 6.2), the manufacturer's standard commercial wiring system shall be furnished. When specified (see 6.2), the wiring shall conform to MIL-C-13486 or SAE J1067 and shall be sized in accordance with the National Electrical Code recommendation for the current capacity required. The cables shall be bound together by wraps placed at 3-inch intervals or inclosed in spiral cut plastic tubing. Grommets shall be furnished to prevent chafing where the cable passes through sheet metal or structural members, and clips shall be used for support. Wiring connectors shall conform to MS27142, MS27143, MS27144, MS27145, MS27146, or MS27147, as applicable. Prior to assembly, connectors shall be lubricated with electrical insulation compound conforming to MIL-S-8660.

3.5.9 Clutch. A hand operated, over-center, engagement-type clutch shall be furnished. The clutch shall transmit not less than 150 percent of the maximum torque developed by the engine. The clutch lining shall conform to SAE J998.

3.5.10 Transmission. When required to meet all rates of application and operations specified herein, a transmission shall be provided and shall be located between the bitumen pump and engine.

3.6 Pump system.

3.6.1 Bitumen pump and piping. The bitumen pump shall be of the positive displacement, rotary type. Suction-line and outlet-line piping shall be provided to extend inlet and outlet of pump, respectively, to accessible locations. This piping shall be not less than 3-inch-inside-diameter pipe, and each pipe shall be provided with faucet end and dust cap components for 3-inch quick-disconnect coupling (see 3.8.3). A perforated, removable, galvanized strainer shall be provided in the suction-line piping to the pump. The pump shall be enclosed in a metal housing equipped with access doors. Provisions shall be made for heating the pump with the portable burner. The housing with access doors shall be constructed and located so that, when

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the portable burner is inserted for heating the pump, the burner flame will not come in contact with, or damage, the pump packing or bearings. The pump shall be equipped with a bypass or other means to prevent damage to the lines when the discharge is closed. The pump drive-shaft coupling shall be secured with a connection that will fail before damage to the pump or transmission occurs. If a shear pin is used, it shall be installed so that when shearing action occurs, shaft alinement is maintained.

3.6.1.1 Type I.

3.6.1.1.1 Pump. The bitumen pump shall be capable of applying bitumen through the spraybar to the ground at a maximum rate which will apply not less than 1.5 gallons per square yard over the full 24-foot width when the distributor is being towed at a speed of not less than 88 feet per minute. Provisions shall be made for flushing and cleaning the pump and spraybar system. The flushing system shall include a supply tank of not less than 20-gallon capacity and all necessary piping. The pump shall supply bitumen to a 24-foot spraybar with sufficient pressure to produce an even, fan-like spray from all nozzles at all rates of application within the distribution range (see 3.6.1.1.5).

3.6.1.1.2 Spraybar and piping. The spraybar shall be full-circulating type and shall consist of a center section and two foldup end sections with provisions to add extensions to each end of the spraybar to make a total spraybar length of 24 feet. The center section with two foldup end sections extended and without extensions shall be 12 feet in length. Extension sections shall be not greater than 2 feet in length. The spraybar shall circulate bitumen through the full 24-foot length without the temperature of bitumen varying more than 10° F throughout the length of the spraybar. Provisions shall be made to fold up the end sections of the spraybar with or without extensions during transit. When folded, the spraybar shall not exceed a width of 96 inches. The spraybar shall lock in this folded position. The elevation of the spraybar nozzle shall not vary by more than $\pm$1/2 inch throughout the length of the 24-foot spraybar. Spraybar supports shall be provided to prevent damage or permanent deformation to the spraybar when operating at any of the spraybar widths. The extended portions of the spraybar shall swing horizontally without damage to the bar or other parts of the distributor when they strike an obstruction while the distributor is moving forward or reverse. The spray bar shall automatically return to its spraying position without interruption of the spraying operation. The bar shall be equipped with a complete set of dripless 1/2-inch nozzles with 1/8-inch slot openings (or with nozzles having equivalent spraying characteristics) and spaced on 4-inch centers. An individual, dripless valve shall be furnished for each nozzle so that flow from any individual nozzle can be stopped without stopping the flow from adjacent nozzle or nozzles. All the nozzle valves shall be linked together so that they can be opened or closed simultaneously by a single-lever operation. The spraybar and section shall fold up or fold back and lock in this position to allow application of bitumen through the center sections only. When an end section, with or without extensions, is folded up or back, as applicable, the nozzle valve linkage shall automatically shut off all valves in the end section, and nozzle valve linkage in the folded end section shall automatically become

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inoperative without manual disconnect. Nozzle valve linkage in the unfolded center section of the spraybar shall remain responsive to single-lever operation. When the folded end section is extended, nozzle valve linkage in the end section shall automatically become responsive to single-lever operation. The spraybar shall be laterally and vertically adjustable by means of mechanical linkages or hydraulic controls operable from the operator's platform. Lateral adjustment from center shall be not less than 8 inches. The vertical adjustment range shall be from 2 inches below normal operating height to raised travel position. The normal operating height shall be established by the supplier and shall be marked so as to be obtainable by the operator without measuring. Provisions shall be made by reverse pumping or equivalent valving to transfer the unsprayed bitumen from the spraybar and piping into a supply tank. All piping and connections for transferring, circulating, spraying, and hand spraying bituminous material shall be furnished. All flexible piping shall be seamless, all-metal, flexible hose made of steel and shall have a single, wirebraid, protective covering. The hose bursting pressure shall be not less than 700 psi. All couplings used with this flexible hose shall be those furnished by the hose manufacturer. The couplings shall be installed as recommended by the hose manufacturer and any hose bending radius shall be not less than that recommended by the hose manufacturer. All valves subjected to pumping pressure shall be of the packed type.

3.6.1.1.3 Controls. All necessary controls for spray application and spraybar circulation shall be accessible to the operator standing on the operator's platform. All other controls necessary to perform the operations specified herein shall be accessible without removal of any housing or parts. All controls shall be clearly identified with permanently affixed identification. Material used for identification marking shall be corrosion resistant. Controls shall be provided to permit the following operations:

a. Pump bitumen from a tank into another tank without bitumen entering the spraybar.
b. Pump bitumen from a tank through the spraybar back to the tank.
c. Pump bitumen from a tank through the spraybar and out the nozzles.
d. Pump bitumen from a tank to the hand-spray attachment.
e. Withdraw bitumen from the hand-spray attachment and return it to a tank.
f. Withdraw bitumen from the spraybar and all piping and return it to a tank.
g. Pump flushing oil through all piping and the spraybar.

3.6.1.1.4 Distance and speed indicating device and distribution chart. The distributor shall be provided with a rubber-tired ground-contact wheel and a cable-driven, tachometer head for recording distance and speed. A housing shall be provided for the tachometer head drive cable. The tachometer head shall provide for the following with a deviation of not more than 5 percent from true distance or speed:

a. A dial indicator to indicate the traveling speed in feet per minute (fpm). The dial shall be calibrated from 0 to 1500 fpm in increments of not more than 20 fpm.

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b. A static pointer which may be manually set to the desired traveling speed.
c. An adjustable linear distance trip recorder reading from 0 to 9,999 feet and resetable to zero.

The ground-contact wheel shall be provided with a screw-type clamping device to permit attaching to the running board of a military dump truck. The tachometer drive cable and housing shall be of sufficient length to permit the tachometer head to be located in the cab of the truck. Provisions shall be made for storing and protecting the ground-contact wheel and the tachometer head and drive cable on the distributor when not in use.

3.6.1.1.5 Distribution. The width of distribution shall be 4 feet, 8 feet, and from 8 feet through the full width of the spraybar in increments of 1 foot. The rates of application shall be from 0.1 gallon through 1.5 gallons per square yard in increments of 0.1 gallon per square yard. The distributor shall apply bitumen through the spraybar to the surface being sprayed at all combinations of widths and rates specified herein. The accuracy of distribution over a given area shall be within ±10 percent. The application at all rates and widths shall be accomplished without streaking or puddling. The forward speed for any distribution rate shall be not less than 88 fpm and shall be not more than 1,350 fpm. Two distribution charts shall be provided with each distributor. The charts shall provide the following for each combination of specified spray widths and spray rates:

a. Metering device reading or setting.
b. Travel speed in fpm.

The charts shall be coated with a clear plastic material for protection.

3.6.1.2 Type II.

3.6.1.2.1 Pump. The bitumen pump shall deliver not less than 350 gallons per minute (gpm) of asphalt cement, grade 85-100 penetration, at a discharge gage pressure of not less than 50 psig. Provisions shall be made for flushing and cleaning the pump.

3.6.1.2.2 Controls. All necessary controls for asphalt transfer operation shall be accessible to the operator standing on the operator's platform. All other controls necessary to perform the operations specified herein shall be accessible without removal of any housing or parts. All controls shall be clearly identified with permanently affixed identification. Material used for identification marking shall be corrosion resistant. Controls shall be provided to permit the following operations:

a. Pump bitumen from one tank to another tank.
b. Reverse the flow of bitumen from that specified in a. above.
c. Pump bitumen to the hand-spray attachment and spray bitumen with the hand-spray attachment.

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d. Withdraw bitumen from the hand-spray attachment and return it to the tank.

3.7 Metering device. A metering device protected from heat shall be furnished. The device shall meter the bitumen either volumetrically or by pressure, as specified (see 6.2). Volumetric metering shall be by means of a pump tachometer calibrated to register in gallons per minute. Pressure metering shall be by means of a pressure gage and suitable controls.

3.8 Accessories. Four transfer hoses, a hand-spray attachment, portable burner, and fittings shall be furnished with each distributor. Provisions shall be made to carry all the accessories on the distributor when they are not in use. Accessories shall remain in their carrying places when the distributor is subjected to any towing or braking requirements specified herein.

3.8.1 Transfer hoses. The four transfer hoses shall each be 8 feet long and of 3-inch flexible hose with high-temperature fittings. Each hose shall be fitted on both ends with hose-end components for 3-inch, quick-disconnect couplings.

3.8.2 Hand-spray attachment. The hand-spray attachment shall consist of a spray tube with nozzle, handle and control cock, cutoff valve, and a half coupling for connection to the hose. The hose shall be 1-inch diameter, 25-feet long, metallic hose with high-temperature fittings. The hose shall be furnished with quick-disconnect couplings on each end. The spray line on the distributor shall be equipped with a faucet-end half coupling for connection to the hose, and with a chain-secured dust cap.

3.8.3 Fittings. All 3-inch, quick-disconnect couplings specified herein shall conform to MS39153. The double adapters for the quick-disconnect couplings shall be furnished.

3.8.4 Portable burner. A portable generating-type burner which burns fuel conforming to VV-F-800, grade DF-1, shall be furnished. The burner shall be used for heating the pump and piping. Three burner at maximum firing rate shall burn not less than 0.5 gallon of fuel per hour and shall have a fuel tank capacity of not less than 4 gallons. The tank shall be marked “FILL WITH DIESEL FUEL ONLY”. Provisions shall be made for carrying the burner on the distributor.

3.8.5 Flush tank. A flushing tank with a capacity of not less than 2 gallons, and including the necessary piping and valving to complete a gravity flow bitumen pump flushing system, shall be provided. The tank shall be located to provide gravity flow to the bitumen pump but positioned to eliminate fire hazard caused by spilling, overflow or drainage of the flushing oil. The tank shall be marked “FILL WITH DIESEL FUEL ONLY”.

3.9 Operator’s platform. An operator’s platform with steps or ladders shall be provided. The platform and steps or ladders shall be made of extended metal to provide an antiskid surface. Steps or ladder rungs shall be not less than 12 inches wide, and there shall be not less than 6 inches of clearance behind ladders to provide toe space.

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3.9.1 Guardrails. Steel pipe guardrails shall be provided around the perimeter of the operator's platform and shall have an opening to provide access to the ground. The guardrail shall be installed at a height of not less than 42 inches above the platform. Each access opening shall be guarded by a removable safety chain.

3.10 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. A lubrication chart in accordance with SAE J753 shall be furnished with each distributor.

3.10.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for qualified products list (QPL) the lubricant supplied shall be from a source that is listed on the applicable QPL (see 6.5).

3.10.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.10.3 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.10.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with either dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain outlets, except outlets fitted with extensions for accessibility, shall be fitted with a magnetic drain plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure when the distributor is parked in its normal service position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a hand held container. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be provided.

3.10.5 Grease lubrication. All grease lubrication points shall be with grease conforming to MIL-G-10924. The distributor shall be assembled, run-in, tested, and delivered with these greases. Each component shall be tagged in a conspicuous place to indicate that military grease has been used.

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3.10.6 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with military oils designated for use in the temperature of 0º to 32° F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.11 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.11.1.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.11.1 Composition. Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required.

3.12 Maintenance.

3.12.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using general purpose tools normally associated with equipment of this nature. After completion of any maintenance operation, the distributor shall operate as specified herein.

3.12.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

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3.13 Safety. All reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces or equipment, including exhaust pipes and other lines which may be subject to high operating temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto the engine, exhaust, burners, or electrical equipment. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Steel walkways and ladders shall be provided where necessary for inspection, servicing, and maintenance. Walkways and ladders shall provide safe and unobstructed access and egress. All platforms, steps, and ladder rungs shall have anti-skid surfaces.

3.14 Noise limits. The noise produced by the distributor shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.2.12. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.15 Slinging attachments. Permanently affixed attachments that enable the distributor to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209, for type II equipment, class 1 or 3 provisions. Slinging attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.16.

3.16 Equipment tiedown attachments. Permanently affixed attachments that permit fastening the distributor to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209, for type II equipment, class 2 or 3 provisions.

3.17 Plates.

3.17.1 Identification and instruction plates. The manufacturer shall provide all identification and instruction plates normally furnished. The plates shall be permanently affixed to the distributor in a conspicuous location.

3.17.2 Shipping data plate. The shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and, in addition, shall show the silhouette of the distributor in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments.

3.17.3 Control identification, information, safety, and warning plates. A plate shall be furnished for each switch, gage, and control, and at points that require special attention for safe

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and efficient operation of the distributor. The plates shall be labeled by lettering or other appropriate marking with information such as function, the direction the control is to be moved, neutral position, safety instructions, and other warnings or instructions as necessary. The plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached as specified in 3.17.1. Plates describing safety measures or warning against operations detrimental to the distributor shall be distinguished by the words "CAUTION" or "WARNING" in red lettering.

3.18 Treatment and painting. The portions of the distributor normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168. The distributor shall then be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns, with the exception that the first article may be camouflage green 383.

3.19 Data for camouflage patterns. The contractor shall generate the following data (see 6.2.1):

a. Eight by ten color photos that mirror each of the views listed under 3.19.1.
b. Original illustrations of each of the views listed under 3.19.1. The illustrations shall be to 1/8 scale on D size paper. Each illustration shall include length, width, and height dimensions relative to each other and shall be clearly detailed to the extent that all discontinuities of the distributor that cover one square inch or more of area, are clearly delineated to scale.

3.19.1 Camouflage pattern views.

a. Back.
b. Front.
c. Left side.
d. Right side.
e. Top.
f. Hidden views that would result in a change of a finished pattern.

Note: Hidden views are those areas requiring camouflage coloration that are not apparent from the other views. Included are variations in the appearance of the distributor that result when the deployed distributor is taken from a stand-by condition and placed into a ready-for-use condition.

3.20 Rustproofing. The distributor shall be rust proofed in accordance with procedures outlined in MIL-R-46164.

3.21 Stenciling. The gross weight of each distributor shall be stenciled on each side of the distributor. "Lift" and "Tiedown", as applicable shall be stenciled adjacent to their respective tiedown and slinging attachment. All stenciling shall be in block-type letters not less than 1 inch high and shall be lusterless black in color.

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3.22 Toolbox. A weatherproof closed compartment or toolbox shall be provided to carry those tools required to operate and maintain the distributor. Toolboxes, when furnished, shall be made of steel not less than 0.0747 inch (USS 14 gage) in thickness. The toolbox shall have a hinged lid and not less than two tension hook-type fasteners that will keep the lid closed when the toolbox is subject to vibration. The toolbox lid shall open at least 90 degrees from the closed position and shall remain in the fully open position either by its own weight or by a holding device. A hasp that will accommodate a padlock with a 3/8-inch shackle shall be provided on the toolbox. The toolbox shall be fastened to the distributor in a protected position. A drain hole shall be provided in the bottom of the toolbox. When a weatherproof compartment is built into the distributor body to carry those tools required to operate the distributor, a portion of the operator's platform may serve as the compartment lid and need not be hinged to the compartment. The compartment lid shall be provided with a hasp that will accommodate a padlock with a 3/8-inch shackle. A drain hole shall be provided in the bottom of the compartment. When specified (see 6.2), the manufacturer's standard toolbox shall be provided.

3.23 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference emission and susceptibility characteristics of the distributor shall meet the recommended limits specified in SAE J55l when measured in accordance with SAE J55l.

3.24 Human factors engineering. The following Society of Automotive Engineers specification shall apply: J1500 (Universal Symbols), J614 (Dipstick Marking), J925 (Access Dimensions), J209 (Instrument Face Design). The distributor shall be designed to be operable and maintainable by 5th through 95 percentile personnel (see SAE J833) who are dressed appropriate to the environments of interest (32° to 125° F). All valves shall be labeled open/close, arrows shall be used to indicate direction of rotation, and all valves shall be labeled with function served by the valve, i.e., "bituminous valve" rather than valve 3. All operator relevant components shall be labeled in terms that communicate to the operator.

3.25 Workmanship.

3.25.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.25.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock-washers shall be provided. All nuts, bolts, and screws shall be tight.

3.25.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet heads shall make full contact with

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the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.25.4 Welding. Surfaces of welds shall be free from imperfections which will be injurious to the weldments. Welding procedures shall be in accordance with a nationally recognized code. The contractor shall make available to the contracting officer certification that each manual welder and automatic welding equipment operator has a current qualification as prescribed by either AWS or ASME code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.25.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defects which reduces the castings ability to perform its intended function.

3.26 Government-furnished property. The following property will be furnished by the Government (see 6.6).

Item No.	Description	Identification	Quantity
1	Patter drawings	Camouflage patterns	As required

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

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b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The distributor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The distributor shall be tested as specified in 4.5.2.1 through 4.5.2.13 for type I or type II, as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each distributor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each distributor shall be tested as specified in 4.5.2.14. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The distributor shall be examined for the following defects:

101.	Dimensions or weight not as specified.	3.1.1
102.	Compatibility not as specified.	3.1.2
103.	Materials not as specified.	3.3
104.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
105.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
106.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
107.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the truck.	3.3.4
108.	Trailer not as specified.	3.4
109.	Frame not as specified.	3.4.3
110.	Suspension not as specified.	3.4.4
111.	Axle not as specified.	3.4.5
112.	Wheels, rims, and tires not as specified.	3.4.6
113.	Fenders and mudguards not as specified.	3.4.7
114.	Rear bumper not as specified.	3.4.8
115.	Intervehicular connections not as specified.	3.4.9
116.	Towbar components not as specified.	3.4.10
117.	Landing gear not as specified.	3.4.11

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118.	Support legs not as specified.	3.4.12
119.	Brake system not as specified.	3.4.13
120.	Chock blocks not as specified.	3.4.14
121.	Vehicular electrical system not as specified.	3.4.15
122.	Engine not as specified.	3.5
123.	Engine instrument panel not as specified.	3.5.1
124.	Engine cranking system not as specified.	3.5.2
125.	Engine lubrication not as specified.	3.5.4
126.	Engine oil sampling valve not as specified.	3.5.5
127.	Engine fuel tank not as specified.	3.5.6
128.	Batteries and batteries box not as specified.	3.5.7
129.	Wiring not as specified.	3.5.8
130.	Clutch not as specified.	3.5.9
131.	Transmission not as specified.	3.5.10
132.	Pump system not as specified.	3.6
133.	Bitumen pump and piping not as specified.	3.6.1
134.	Metering device not as specified.	3.7
135.	Accessories not as specified.	3.8
136.	Operator's platform not as specified.	3.9
137.	Guardrails not as specified.	3.9.1
138.	Lubrication not as specified.	3.10
139.	Lubrication fittings not as specified.	3.10.2
140.	Pressure-release device not as specified.	3.10.3
141.	Filling, draining and checking provisions not as specified.	3.10.4
142.	Fungus and moisture resistance not as specified.	3.11
143.	Ease of maintenance not as specified.	3.12
144.	Interchangeability not as specified.	3.12.2
145.	Safety not as specified.	3.13
146.	Noise not as specified.	3.14
147.	Slinging attachments not as specified.	3.15
148.	Equipment tiedown attachments not as specified.	3.16
149.	Plates not as specified.	3.17
150.	Treatment and painting not as specified.	3.18
151.	Color not as specified.	3.18
152.	Camouflage pattern data not as specified.	3.20
153.	Camouflage patterns not as specified.	3.20
154.	Rustproofing not as specified.	3.19
155.	Stenciling not as specified.	3.21
156.	Toolbox not as specified.	3.22
157.	Electromagnetic interference not as specified.	3.23
158.	Human factors engineering not as specified.	3.24
159.	Steel fabrication not as specified.	3.25.1
160.	Bolted connections not as specified.	3.25.2
161.	Riveted connections not as specified.	3.25.3
162.	Welding not as specified.	3.25.4
163.	Castings not as specified.	3.25.5

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3.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The distributor shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Portable burner. Start the burner and fire at maximum rate. Insert the burner into the pump housing and observe the impingement of the flame. Determine the rate of fuel consumption for a period of not less than 15 minutes. Impingement of the flame on the pump packing or bearings, or fuel consumption of less than 0.5 gallon per hour shall constitute failure of this test.

4.5.2.3 Commercial engine. During the distributor operational test specified in 4.5.2.8 and 4.5.2.10, the commercial engine speed and intake manifold vacuum shall be measured and recorded at 30-minute intervals for the duration of the test. These readings shall be compared with the engine speed and horsepower ratings published by the engine manufacturer. Any intake manifold vacuum reading recorded during the distributor operational test that is lower than the reading listed on the engine manufacturer's published continuous horsepower rating, or 90 percent of the engine manufacturer’s published maximum horsepower rating, at the applicable speeds (rpm) shall constitute failure of this test.

4.5.2.4 Slinging attachments.

4.5.2.4.1 Test procedures.

a. Lift the distributor and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the distributor. Determine the forward angles of application for each attachment. Measure the clearance between each sling and the distributor. Measure eye openings and clearance dimensions of the attachments.
b. Restrain the distributor by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in a. above. An alternate method of test is to lift the distributor as in a. above and add weights to the main frame until the required force is obtained. Hold the distributor under either of the above loading conditions for at least 90 seconds.

4.5.2.4.2 Failure criteria. Weld failure or permanent deformation caused by lifting or failure to meet requirements of MIL-STD-209, shall constitute failure of this test.

4.5.2.5 Equipment attachments.

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4.5.2.5.1 Test procedure. With the distributor anchored by means other than the attachment being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedowns.

4.5.2.5.2 Failure criteria. Failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.2.6 Road. The following road test shall be performed. Couple the trailer-mounted distributor to a commercial design dump truck compatible to a 5 ton, 6 x 6, military design dump truck, and tow the distributor as specified in table IV, with the full complement of attachments, and accessories stowed for travel.

TABLE IV. Road test.

Course	Mileage and speed
Primary highway Secondary road	200 miles at variable speeds up to 55 mph. 100 miles at variable speeds up to 30 mph

During the road test, perform and record the actual results of the following:

a. Tow the trailer-mounted distributor on side slopes up to 10 percent, having the curb side downhill and having the road side downhill. Make full-throttle starts and panic stops on the side slopes.
b. Tow the trailer-mounted distributor over 5-inch-high obstacles such that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure for evidence of misalinement.
c. Tow the trailer-mounted distributor over V-ditches not less than 36 inches wide and 12 inches deep. Observe ground contact areas for possible permanent deformation other than paint. Negotiate the ditches both perpendicular and at an angle of 45 degrees to the direction of travel, until at least 10 V-ditch crossings have been completed.
d. Measure and record the deviation or off tracking of the trailer with respect to the prime mover, when the trailer-mounted distributor is towed straight ahead on a paved surface.
e. Jackknife the trailer-mounted distributor at angles up to 60 degrees from straight ahead both to the left and to the right.
f. Turn the prime mover, towing the trailer-mounted distributor, in a rolling, sharpest-turning circle for two complete turns and repeat in the opposite direction.
g. Turn the trailer lights on, and immerse or back the trailer into water not less than 20 inches deep for not less than 2 minutes. Stop the trailer using the trailer brakes and turn off the lights for not less than 10 minutes. Turn on the lights. Tow the trailer out of the water and make a brake stop using the trailer brakes and observe

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trailer braking for water fade, grabbing, and chatter. Observe all brake system components and electrical system components, except hermetically sealed lights, and inspect at least the hubs and bearings of one wheel for damage.

4.5.2.6.1 Failure criteria. Failure to complete the test, inability to operate and function as specified herein, accessories damaged or not remaining in their carrying places shall constitute failure of this test.

4.5.2.7 Brake. Test the trailer for conformance to DOT Motor Carrier Regulations, the California Vehicle Code, and to the requirements specified herein:

a. Tow the trailer-mounted distributor over the test course and when the trailer reaches a speed of not less than 20 mph, make a panic brake stop using both prime mover and trailer brakes. Perform the test four times, allowing not more than 3 minutes between test.
b. Tow the trailer-mounted distributor at not less than 50 mph and apply service brakes to produce a deceleration rate of 10 feet per second, until the trailer has slowed to 30 mph. Then immediately accelerate at wide-open-throttle again up to not less than 50 mph. Repeat the snubbing cycle for a total of 5 cycles and make a panic stop on the 6th cycle.
c. Tow the trailer-mounted distributor over the test course and, when the trailer reaches 20 mph, cause the trailer to go into "emergency", simulating a trailer breakaway from the prime mover. Restore the trailer brake system to normal. Disconnect the trailer emergency line hose and observe if there is an air leak at the trailer gladhand.
d. Release the emergency-parking brakes using only tools supplied with each distributor or using a slave air source.
e. Test the trailer-mounted distributor brake system as specified in TTMA RP No. 12.

4.5.2.7.1 Failure criteria. Failure to complete the test, inability to function and operate as specified or nonconformance to DOT regulations, the California Vehicle Code, or TTMA requirements shall constitute failure of this test.

4.5.2.8 Type I operational and distribution.

4.5.2.8.1 Spraybar circulation. With a bitumen supply tank filled with asphalt cement, grade 85-100 penetration, heat to a temperature of not less than 300° F. Circulate the asphalt cement from a tank through the 24-foot spraybar and back into a tank for a period of not less than 15 minutes. Record the temperature of the asphalt cement at the center and at both ends of the bar after the circulation period. A temperature variation of more than 10° F throughout the length of the spraybar shall constitute failure of this test.

4.5.2.8.2 Pump capacity. Spray asphalt cement, grade 85-100 penetration, at a temperature of not greater than 285° F through the full 24-foot-width spraybar into a container for 60 seconds at maximum spray rate. The 60-second time period shall be determined by a stopwatch, and the

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volume of asphalt cement shall be determined by weight using unit weight and weight-to-volume conversion. A volume of asphalt cement of less than 350 gallons sprayed into the container during this 60-second period shall constitute failure of this test.

4.5.2.8.3 Application and distribution. Install the distance and speed indicating device as specified on a prime mover with a supply tank, tow the distributor, and apply 85-100 penetration asphalt cement through the 24-foot spraybar at rates of 0.4, 0.8, and 1.5 gallon(s) per square yard over a predetermined distance of not less than 100 feet for each rate. The 1.5 gallons per square yard rate shall be applied while going down a grade. Then apply 85-100 penetration asphalt cement through 4 feet of spraybar at a rate of 0.1 gallon per square yard over a distance of not less than 100 feet while going up a grade. The following procedures shall be followed for testing at each of these rates:

a. Weigh the tank filled with bitumen.
b. Apply the bitumen over the predetermined distance at the required rate of application (speed of truck and metering device reading or setting should be obtained from distribution chart specified herein).
c. Using a stopwatch, determine time required for the distributor to apply bitumen over the predetermined distance.
d. Weigh the tank after the bitumen has been applied.
e. Determine amount of bitumen applied by difference in weight of the tank before and after test and convert to gallons (use applicable certified specific gravity versus temperature curve).
f. Compute rate of application in gallons per square yard by utilizing amount of bitumen applied, travel distance, application time, and spraybar width.

Observe spray pattern during all rates of application and observe applied bitumen after each application for streaking or puddling. Examine the nozzles for drip after shutoff. With the spraybar at maximum width, operate the distributor at the speed required for an application rate of 0.1 gallon per square yard, and bring one end of the spraybar into contact with a rigid obstacle. Observe the horizontal swing breakaway feature. Individually shut off one nozzle in the center section of the spraybar to demonstrate individual shutoff feature. Then fold and lock the 24-foot spraybar and apply bitumen through the center section only. Any of the following shall constitute failure of this test: rate of bitumen actually applied not within +10 percent of the predetermined rate; spray from any nozzle not extending to the centerline of adjacent nozzles; streaking or puddling of applied bitumen; the extended end section of the spraybar not swinging horizontally without damage to the spraybar or other parts of the distributor when extended section strikes a rigid obstacle; the extended end sections of the spraybar not folding or locking in folded position as specified; extended end section valve linkage not automatically disconnected, or nozzle valves not closing and remaining in closed position when extended section of the spraybar is locked in folded position; spraybar valve linkage not operating as required when spraybar is either folded or extended; nozzles dripping after shutoff; flow from individual nozzle not being shut off.

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4.5.2.8.4 Operations. The following operations shall be performed:

a. Pump bitumen from a tank into another tank without bitumen entering the spraybar.
b. Pump bitumen from a tank through the spraybar back to the tank.
c. Pump bitumen from a tank through the spraybar and out the nozzles.
d. Pump bitumen from a tank to the hand-spray attachment. Spray bitumen using hand-spray attachment.
e. Withdraw bitumen from the spraybar and all piping and return it to a tank.
f. Pump flushing oil through all piping and the spraybar.
g. Withdraw bitumen from the hand-spray attachment and return it to a tank.

4.5.2.8.5 Spraybar adjustments and alinement. Adjust the spraybar laterally and vertically throughout the full range for each adjustment and measure the range of adjustment. Then adjust the 24-foot spraybar for no crown and measure the elevation of nozzles at the center and both ends of the spraybar. Fold up the extended ends of the 24-foot spraybar into its travel position and measure the spraybar width. Spraybar lateral adjustment or vertical adjustment less than specified, elevation of any nozzles not within $\pm$1/2 inch of any other nozzle, spraybar not folded up and locked as specified, or width of spraybar in travel position greater than 96 inches shall constitute failure of this test.

4.5.2.8.6 Failure criteria. Inability of the distributor to perform any of the above specified operations shall constitute failure of this test.

4.5.2.9 Speed and indicating device. Tow the distributor over a measured distance of not less than 5,000 feet at an indicated speed of 500 fpm. Determine by a stopwatch and record the time required to travel the measured distance at the indicated speed of 500 fpm. Record the trip-indicator reading at the start and at the end of the measured distance traveled.

4.5.2.9.1 Failure criteria. Indicated speed not being within 5 percent of actual speed or indicated distance traveled not being within 5 percent of actual distance traveled shall constitute failure of this test.

4.5.2.10 Type II operational.

4.5.2.10.1 Pump capacity. Pump asphalt cement, grade 85-100 penetration, at a temperature of not greater than 285° F into a container for 60 seconds at a discharge pressure of not less than 50 psig and at maximum discharge rate. The 60-second period shall be determined by a stopwatch, and the volume of asphalt cement shall be determined by weight using unit weight and weight-to-volume conversion.

4.5.2.10.2 Operations. The following operations shall be performed:

a. Pump bitumen from one tank to another tank.
b. Reverse the flow of bitumen from that specified in a. above.
c. Pump bitumen to the hand-spray attachment and spray bitumen with the hand-spray attachment.
d. Withdraw bitumen from the hand-spray attachment and return it to a tank.

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4.5.2.10.3 Failure criteria. A volume of asphalt cement of less than 350 gallons discharged into the container during this 60-second period shall constitute failure of this test. Inability of the distributor to perform any of the above specified operations shall constitute failure of this test.

4.5.2.11 Performance. Operate the distributor using high-viscosity fuel oil for a run-in period of not less than 60 minutes. When a diesel engine is furnished, this test shall be performed using the diesel fuel specified in 3.5.3. Check the distributor for leaks, demonstrate operation of all controls, and make necessary adjustment prior to shipment. Any leak, inability of any control to operate as required, or malfunction of any component that is not corrected by adjustment shall constitute failure of this test.

4.5.2.12 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) ARC increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Exterior noise levels for drive-by shall not exceed 88dB(A) when tested in accordance with SAE J88. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.12.1 Failure criteria. Failure to reduce the noise level or provide documentation verifying that compliance is either technically or economically unfeasible shall constitute failure of this test.

4.5.2.13 Human factors engineering inspection. The distributor shall be inspected for conformance with 3.24 requirements. Failure to comply shall constitute failure of this inspection.

4.5.2.14 Electromagnetic interference. When specified (see 6.2), the distributor shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article test reports.

4.5.2.14.1 Failure criteria. Failure to meet the requirements of SAE J551, shall constitute failure of this test.

4.5.2.15 Quality conformance operational run-in. Each distributor shall be serviced as required and shall be run-in in accordance with the manufacturer's normal procedures for a period of not less than 30 minutes.

4.5.2.15.1 Failure criteria. One or more functional defects shall constitute failure, and acceptance of the equipment shall not be approved until all defects are corrected.

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4.6 Inspection comparison. The Government may select distributors at any time during the contract production period and subject these distributors to the examination specified in 4.5.1 and the tests specified in 4.5.2 through 4.5.2.13 to determine that the quality of the selected distributors is equal to the quality standards established during first article inspection. The inspection will be performed by the Government at a site selected by the Government. Distributors will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article distributors.

4.6.1 Inspection failure. Failure of any distributor when inspected as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of distributors until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.7 Inspection of packaging. The preservation, packing, and marking for level A or level B shall be examined to determine compliance with the quality assurance provisions of MIL-D-771. Commercial preservation, packing and marking shall be examined to determine compliance with ASTM D 3951.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete distributor shall be preserved, packed, and marked in accordance with MIL-D-771 for level A or level B packaging and in accordance with ASTM D 3951 for commercial packaging. The degree of preservation and packing shall be level A, level B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The distributor is intended for use in transferring bitumen from one tank to another and for accurately applying bitumen to roadway surfaces.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of distributor required (see 1.2).
c. When first article inspection is not required (see 3.2).
d. Time frame required for submission of the first article (see 3.2).
e. When a spare tire carrier is required (see 3.4.6.2).
f. When a spare tire and wheel assembly is required (see 3.4.6.3).
g. When a vehicular 12-volt electrical system shall be provided (see 3.4.15).
h. When a 24-volt lighting system shall be provided (see 3.4.15.1).
i. Type of engine required (see 3.5).
j. When the engine cranking system shall be other than as specified (see 3.5.2).
k. When the batteries shall be other than as specified (see 3.5.7.1).

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l. When other than the manufacturer's standard commercial wiring shall be used (see 3.5.8).
m. Whether a volumetric or pressure metering device is required (see 3.7).
n. When a fungus and moisture resistance treatment is not required (see 3.11).
o. When treatment and painting shall be other than as specified (see 3.18).
p. When color shall be other than specified (see 3.18).
q. Time frame when camouflage pattern data is required (see 3.19).
r. When manufacturer's standard toolbox shall be provided (see 3.22).
s. When electromagnetic interference is not required (see 3.23).
t. Degree of preservation and degree of packing required (see 5.1).

6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, contract data requirements list (CDRL), the data requirements identified below shall be developed as specified by an approved data item description (DD Form 1644) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements	Applicable DID No.
a. 3.19	Camouflage Pattern Photos	UDI-V-24003B/M (see figure 2)
b. 3.19	Camouflage Pattern Illustrations	DI-M-30420/M (see figure 3)

6.3 First article. When a first article inspection is required, the items should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research & Development Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under chapter 2 of MIL-HDBK-113.

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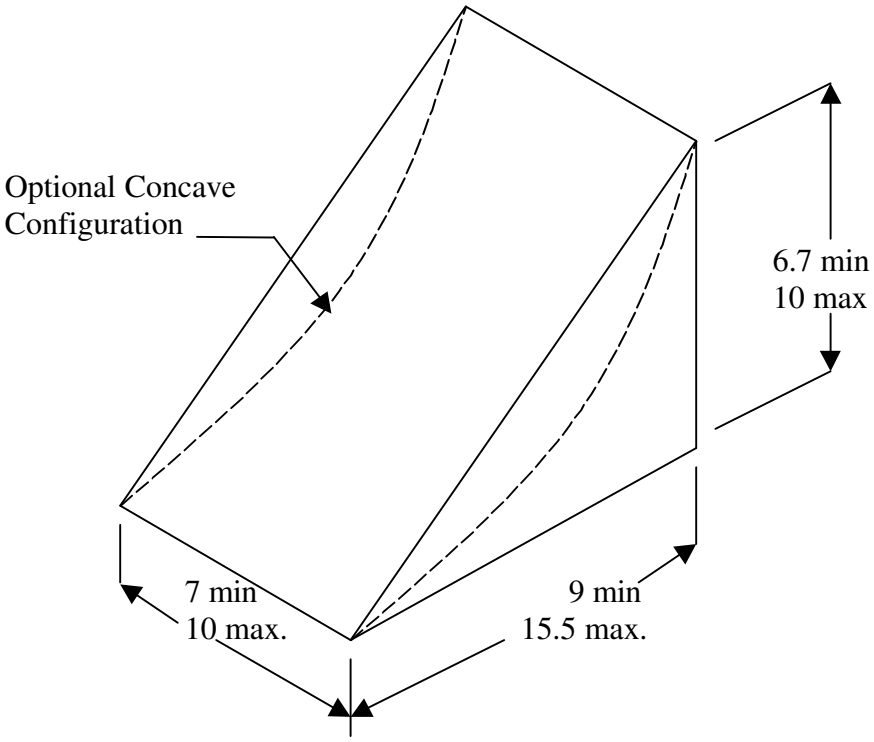
6.6 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.26.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 3895-A223

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT-11.5 LB

CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 1. Chock block assembly.

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DATA ITEM DESCRIPTION

2. IDENTIFICATION NO(S).	
AGENCY	NUMBER
NAVY-YD	UDI-V-24003B/M

1. TITLE
PHOTOGRAPHIC PRINTS AND SLIDES

3. DESCRIPTION/PURPOSE
This data consists of photographic prints and/or slides which provide a pictorial medium for exchange of technical information between developing Agency and users and becomes a permanent historical and legal record by recording configuration during the course of program development.

4. APPROVAL DATE
13 June 1972

5. OFFICE OF PRIMARY RESPONSIBILITY
NAVFAC 0434

6. DDC REQUIRED

8. APPROVAL LIMITATION

7. APPLICATION/INTERRELATIONSHIP
7.1 Deliverable data covered by this DID and specified in block 16 of the CDRL, is required as a visual record of equipment procured under contract.

9. REFERENCES (Mandatory as cited in block 10)

MCSL NUMBER(S)

10. PREPARATION INSTRUCTIONS
10.1 These photographs shall be standard 8x10 inch prints and furnished in 2x2 inch cardboard frames; and unless otherwise specified on the DD Form 1423. Views shall include, but not be limited to, a front, side, and rear view. Background of photographs shall be void of unrelated objects.

10.2 Interior views shall be in sufficient quantity to show arrangement and layout of installed equipment.

10.3 Photographic prints shall show excellent detail and shall be color suitable for reproduction.

DD FORM 1664

FIGURE 2. Camouflage pattern photos.

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DATA ITEM DESCRIPTION	2. IDENTIFICATION NO(S).	
	AGENCY	NUMBER
1. TITLE Technical Illustrations and Graphs	USAF	DI-M-30420/M
3. DESCRIPTION/PURPOSE To provide wave plotting, drafting, computations, data reduction and/or mathematical analysis in support of research projects.	4. APPROVAL DATE 29 Oct 1976	
	5. OFFICE OF PRIMARY RESPONSIBILITY AFSC	
	6. DDC REQUIRED	
	8. APPROVAL LIMITATION	
7. APPLICATION/INTERRELATIONSHIP Mathematical Services Projects	9. REFERENCES (Mandatory as cited in Block 10)	
Replaces (U)M-177/ESD	MCSL NUMBER(S)	

10. PREPARATION INSTRUCTIONS

Data will encompass plotting, drafting and artistic creative illustrations of apparatus and devices. The work may include three-dimensional pictorial representations. Completed reproducible art work must be of sufficiently high quality to be used for publication in technical journals and for visual aids at conferences, reviews and briefings.

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FIGURE 3. Camouflage pattern illustrations.

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